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Review

Liquid scintillation detectors for high energy neutrinos

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ABSTRACT

Large liquid scintillation detectors have been generally used for low energy neutrino measurements, in the MeV energy region. We describe the potential employment of large detectors (>1 kiloton) for studies of higher energy neutrino interactions, such as cosmic rays and long baseline experiments. When considering the physics potential of new large instruments the possibility of doing useful measurements with higher energy neutrino interactions has been overlooked. Here we take into account Fermat's principle, which states that the first light to reach each PMT will follow the shortest path between that PMT and the point of origin. We describe the geometry of this process, and the resulting wavefront, which we call the "Fermat surface", and discuss methods of using this surface to extract directional track information and particle identification. This capability may be demonstrated in the new long baseline neutrino beam from Jaeri accelerator to the KamLAND detector in Japan. Other exciting applications include the use of Hanohano as a movable long baseline detector in this same beam, and LENA in Europe for future long baseline neutrino beams from CERN. Also, this methodology opens up the question as to whether a large liquid scintillator detector should be given consideration for use in a future long baseline experiment from Fermilab to the DUSEL underground laboratory at Homestake.

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1. Introduction

A single muon moves horizontally ($+x$ direction) through the center of a cylindrical detector, 40 m tall by 40 m in diameter. The muon has an energy of about 1GeV, making the track length approximately 5 m (considered to start at $x = -2.5$ m so that it is centered in the detector). As the muon travels through the scintillator, it produces light isotropically from every point along the track. This can be imagined to be like incoherent Huygens wavelets, adding up to create a three dimensional wavefront. In the forward direction, this looks very much like a Cherenkov cone (though unpolarized) expanding out in y and z while moving to the right, in the $+x$ direction. In addition to this, there is also a spherically symmetric wavefront expanding to the left, centered at the start of the track. And also, to the right of the Cherenkov cone, there is a spherical wavefront centered at the end of the track.

In this way, you can fit all the first hit times to two point fits and a Cherenkov cone. We call this three dimensional wavefront the Fermat surface, since Fermat's principle illustrates that no light from the track will reach the PMTs faster than the light along this wavefront. This is demonstrated in Fig. 1.

There is plenty of information to work with; in a detector of this size there will be a few thousand illuminated PMTs, each with a time and total charge. The first hits will be closest to the Fermat surface time. Full waveform records may also be possible in newer detectors. Thus the fit is greatly over constrained.

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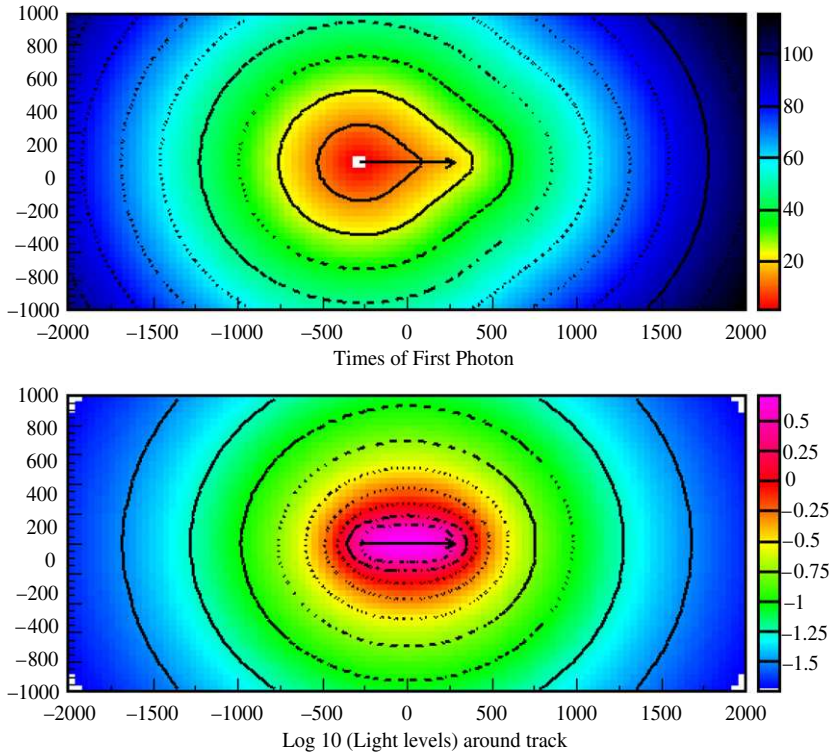


Fig. 1. Contours of first hit times (upper) and total light level for a 547 cm muon track in liquid scintillator. The asymmetric time contours point back to the start of the track, while the amplitudes give the center of the track, together uniquely identifying the track length and the position.

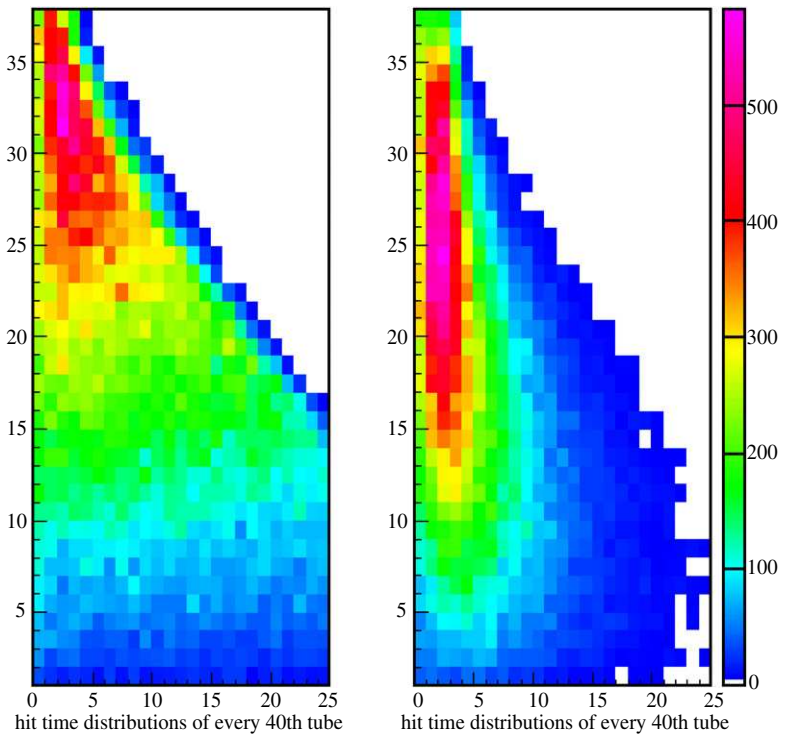


Fig. 2. Time distributions for a 1 GeV muon track (left) and similar electron shower (right). Hit times (horizontal axis) from a selection of 40 phototubes (vertical axis) on the top of the 40 m × 40 m cylindrical detector are shown. Times are relative to the first photon for that phototube.

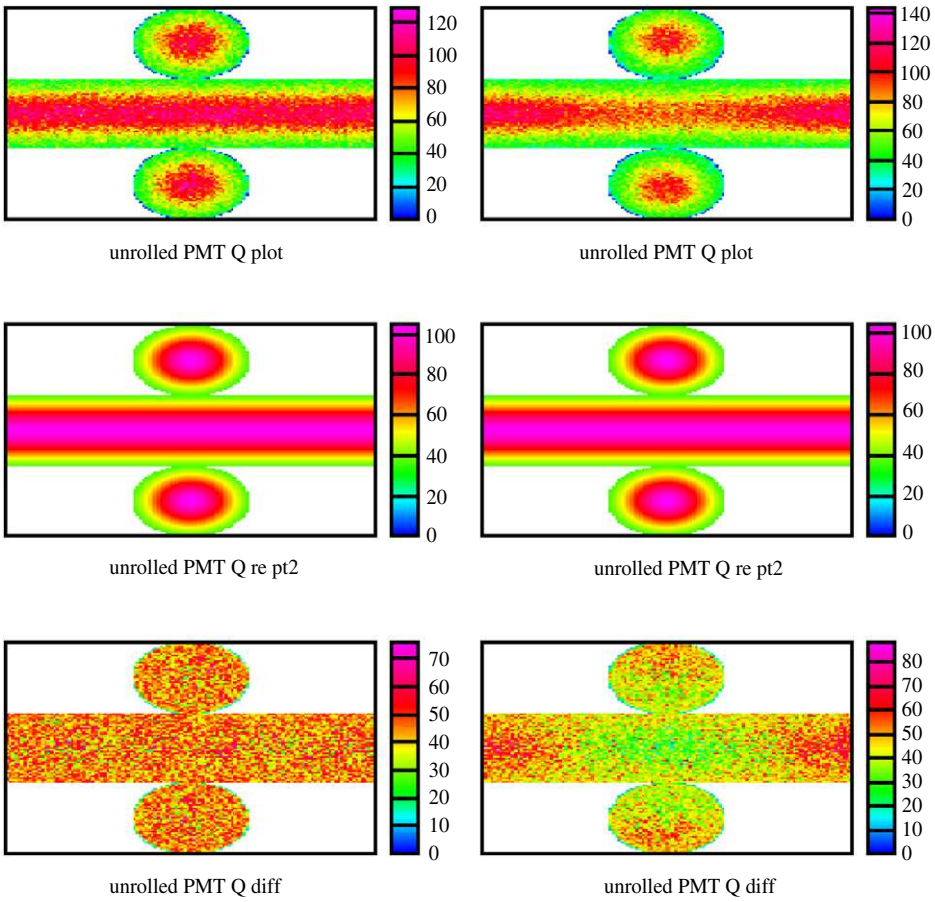


Fig. 3. Comparison of the first arrival times for a muon track (left) and electron shower (right), for 1 GeV events in both cases. The residuals between simulated data (top) and point fits to the interaction vertex (middle) are shown in the bottom panel. These panels are modelled on the SuperKamiokande "unrolled view", with the top of the detector folded up, bottom down, and the barrel of the cylinder unrolled.

This was explored with a simple Monte Carlo simulation of a right cylindrical detector, 40 m tall and 40 m in diameter, or 50, 265 m^3. It is filled with next generation liquid scintillator (doped linear alkylbenzene, LAB), producing on average 10,000 detectable photons/MeV. It is taken to have 7377 phototubes each with a collection area of 0.4 m^2, mounted on a 102 cm lattice. The simulation ignores scattering and attenuation, stepping the muon along at the speed of light in vacuum while radiating 20,000 photos per cm in random directions. Each photon is tracked toward the detector wall at the speed of light in vacuum divided by the index of refraction of the medium ($n = 1.482$). A resolution of 2 ns is inserted. The time of the first hit in, the total number of hits and the median time are saved for each PMT. The generated data is shown in Fig. 1.

There is also some information to be extracted from the multiple hits at each PMT. Fig. 2 shows that there are visible differences between the photon arrival times for a muon track and for the same energy electron shower.

2. Event topology

The first hit times at each PMT are shown in the top panel of Fig. 3. The middle panel shows a point source distribution of first hit times emanating from the beginning of the muon track. The bottom panel shows the difference between these two, demonstrating that a point fit on the first hit time data reveals the origin of the track fairly well. On the right side, the same things are presented for a 1 GeV electron shower, showing that particle identification may be possible.

A fit to the total charge deposited at each PMT gives a nice determination of the track center, as demonstrated in Fig. 4. The simulated muon goes through the detector center and is pointed toward the back. The top panel shows the charge collected by each PMT, the second panel shows the charge expected from a point source of radiation of the same total amount of light, but radiating from the center of the track. The third panel shows the difference between these two, and as one sees the point fit gives a very nice reproduction, indicating a simple means to start track finding. This charge point fit gives a good measure of the track center.

The resolutions of a set of time based point fits are shown in the left hand panel of Fig. 5. This method finds the track origin to within 10 cm (though systematically shifted by 80 cm). This is a rudimentary start to track finding. A third fitting

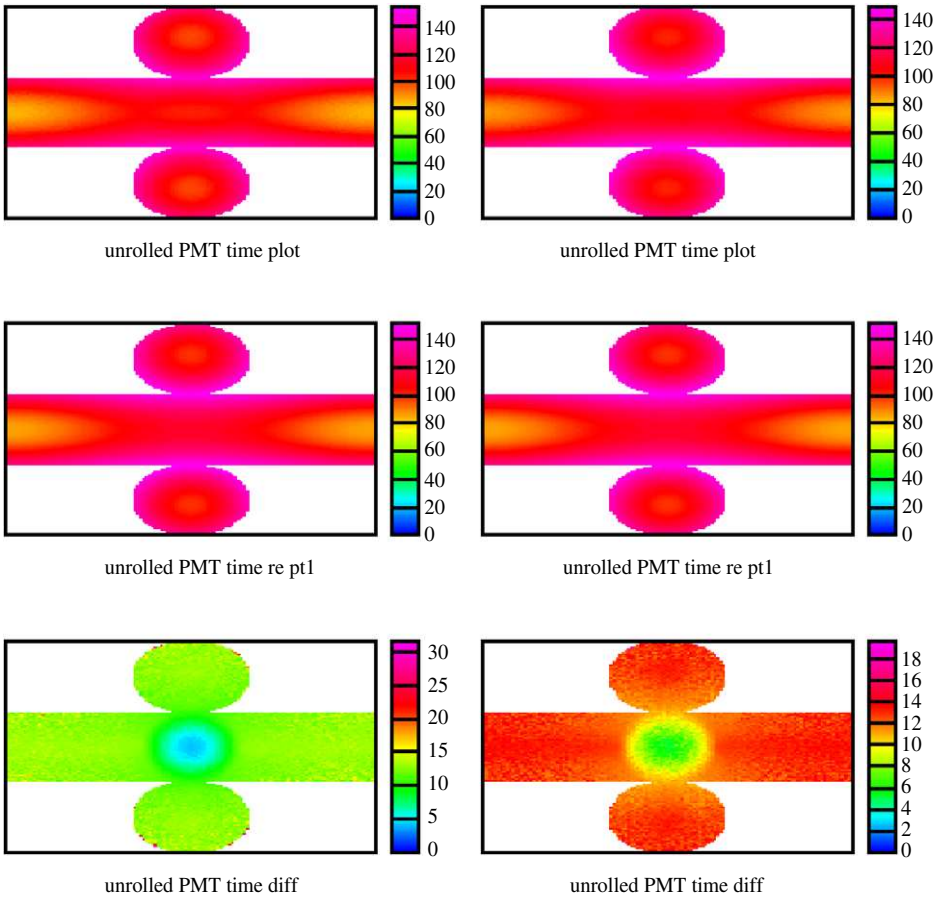


Fig. 4. A comparison similar to Fig. 3 is made for the total charge. The simulated data is shown at the top, a point fit to the vertex in the middle, and the difference is taken at the bottom. Again, the muon track is on the left and the electron shower to the right.

was done to the Fermat timing of a line (pseudo-muon). The time from the start of the track to a point given by the vector $\mathbf{R}_p$ is:

$$ct = |\mathbf{R}_p|[\sin(\alpha)\tan(\theta_c) + \cos(\alpha)]$$

and the angle, α, between track and PMT directions is given by:

$$\cos(\alpha) = \mathbf{R}_p\mathbf{R}_t/|\mathbf{R}_p|$$

where the unit vector R_t points along the track.

The two right panels of Fig. 5 show a χ^2 scan as a test track is moved along the 'real' track and as the test track is varied in angle with respect to the real track. The resolution in longitudinal fitting is about 1 cm, and in angle about 0.2 degrees. These results might be a bit optimistic, but they are at least competitive with, if not even better than those for SuperK. (The SuperK-I single ring resolution for electrons was 3 degrees, and 1.8 degrees for muons [1].)

Since the original work reported here [2], two groups have verified and extended the projected capabilities of large liquid scintillation detectors [3,4]. All groups agree that muon and electron quasi-elastic events should be recognized with very high reliability.

We are also working on methods to visualize tracks, without making assumptions on event topology, drawing upon experience with medical imaging. While this work is not mature, it shows promise for the reconstruction of complex events.

3. Applications

This type of detector may have some significant applications, beyond the studies of geoneutrinos, reactor neutrinos, relic supernova, and remote reactor monitoring. With directionality and particle identification, the possibilities for long baseline experiments with liquid scintillator detectors is encouraging. Tokai to Kamioka, or T2K, is a good candidate, with KamLAND already situated in the beam path [5–7].

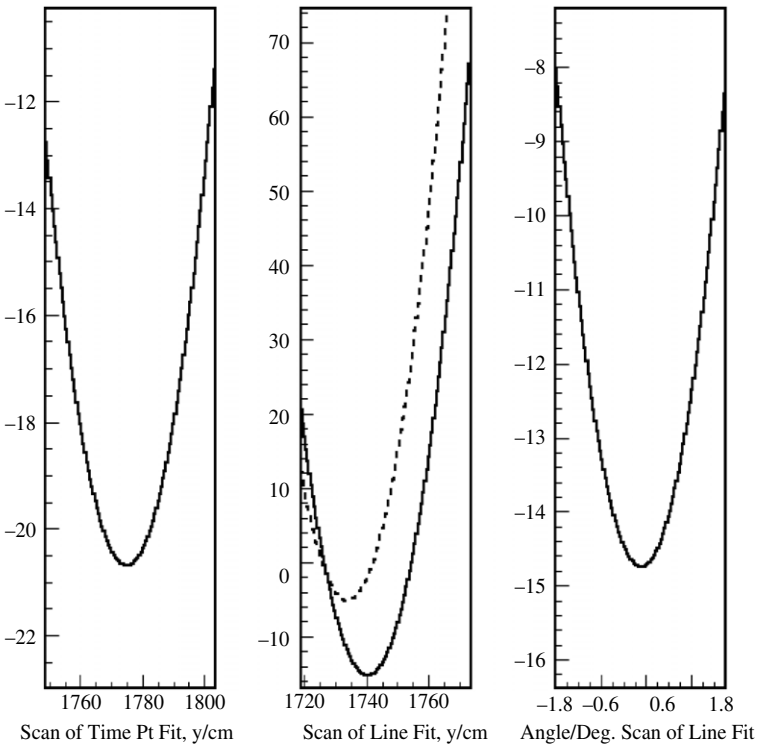


Fig. 5. Point fit to time (left) resolves to within 1 cm of start of track, and angular resolution (right) is 0.2 degrees. Middle panel shows comparison of line fit to muon and electron events, with the electron about 11 σ worse fit.

Another potential long baseline experiment could be done with Hanohano [8–10]. As a portable, ocean deployed detector, it could easily be placed in the Tokai beam, or other, future beams. At 10 kiloton, and an increased distance, we would get about 1/8 the rate as SuperK [1]. Although Hanohano-like designs of up to 100 kiloton are feasible from an engineering standpoint [8]. Similarly the proposed LENA detector in Europe, at 50 kiloton, may now be considered as an end station for future CERN long baseline neutrino beams [11].

For energies in the same 1 GeV range we are talking about, a large liquid scintillation instrument would provide very nice detection of neutrinos from low mass WIMP annihilation in the sun. This Fermat surface technique might also improve the prospects for observation of nucleon decay, or astrophysical sources in the GeV range.

It should be noted that aside from the cost of liquid scintillator, the upper size limit of such modules will be determined by the attenuation length for light in scintillator, which is considerably less than in pure water. These combined effects may limit such detectors to less than 100 kilotons, whereas for megaton scale detectors we surely will need water based instruments.

4. Summary

The use of a large liquid scintillation instrument for similar physics to that which can be done with a large water Cherenkov detector has been sketched out. Such a detector may equal or even outperform the water Cherenkov instruments at the GeV scale. Liquid scintillation based instruments have the added benefit of simultaneously being able to explore the lower energy range of electron anti-neutrinos due to reactors and geological origins, which are inaccessible to a water based Cherenkov detector. At the very least, this new avenue of physics for liquid scintillation experiments deserves further exploration.

Acknowledgements

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